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Project(s) related: E-Learning System Ontology

Ontology Requirements Specification Document	
1	**Purpose**
	This ontology is designed to model and organize the main entities and semantic relationships in an e-learning system. The ontology supports structured representation of academic activities such as enrollment, teaching, assignment management, submission tracking, grading, semantic reasoning, and educational analytics. The ontology also supports future ontology population using Large Language Models (LLMs) and intelligent reasoning mechanisms for advanced educational applications.
2	**Scope**
	This ontology focuses on an E-Learning system including students, instructors, courses, modules, assignments, submissions, grades, learning objectives, and feedback processes. **Included:** • Students, Instructors, Courses, Modules, Assignments, Submissions, Grades • Learning Objectives and Feedback • Enrollment and teaching relationships • Assignment submission and evaluation • Semantic reasoning and prerequisite relationships • Ontology population concepts using LLMs **Excluded:** • Authentication systems (login/password management) • Communication systems (chat/forums) • Video streaming and storage infrastructure • Financial/payment systems • Advanced production-level analytics platforms
3	**Implementation Language**
	The ontology is implemented in OWL/RDF and developed using Protégé. Consistency validation and semantic reasoning were performed using the HermiT reasoner.

4	**Intended End-Users**
	The intended users of this ontology include: • Students • Instructors • Academic administrators • Educational researchers • Semantic Web developers • E-learning platform designers
5	**Intended Uses**
	This ontology is intended to: • Organize academic knowledge semantically within an e-learning environment • Support structured semantic querying using SPARQL • Represent educational workflows and relationships • Support ontology reasoning and consistency validation • Enable future ontology population using LLMs • Serve as a foundation for intelligent educational systems and recommendation engines
6	**Ontology Requirements**
	a. Non-Functional Requirements
	• NFR1: The ontology shall be implemented in OWL/RDF using Protégé. • NFR2: The ontology shall remain logically consistent under OWL reasoning. • NFR3: The class hierarchy shall be modular, reusable, and extensible. • NFR4: All object and datatype property domains/ranges shall reference valid ontology classes. • NFR5: Semantic reasoning shall support transitive prerequisite inference. • NFR6: The ontology shall support future ontology population pipelines. • NFR7: The ontology shall exclude authentication and infrastructure-specific systems.
	b. Functional Requirements
	• FR1: The ontology shall identify students enrolled in specific courses. • FR2: The ontology shall identify instructors teaching courses. • FR3: The ontology shall manage assignments, submissions, grades, and feedback. • FR4: The ontology shall support prerequisite reasoning between courses. • FR5: The ontology shall support semantic classification of late submissions. • FR6: The ontology shall associate learning objectives with courses and modules. • FR7: The ontology shall support future AI-assisted educational analytics.
7	**Competency Questions**
	Core Competency Questions CQ1: Which students are enrolled in a specific course? CQ2: Which instructor teaches a specific course?

	CQ3: Which assignments belong to a specific course? CQ4: Which students submitted a specific assignment? CQ5: What grade did a student receive for an assignment? CQ6: Which courses have prerequisites? CQ7: Which assignments have a deadline? CQ8: Which modules belong to a specific course? CQ9: What is the submission date for a student's submission? CQ10: Which courses does a specific instructor teach? **Extended Competency Questions** CQ11: If Course A requires Course B, and Course B requires Course C, does Course A require Course C? CQ12: Can a student be enrolled in multiple courses at the same time? CQ13: Does each submission have exactly one grade? CQ14: What is the average score across all submissions for a given course? CQ15: Which students submitted an assignment after its deadline? CQ16: What learning objectives are associated with a course or module? CQ17: What qualitative feedback was provided for a specific submission? CQ18: Which submissions were automatically classified as LateSubmission instances?
8	**Pre-Glossary of Terms**
	Terms from Competency Questions: Student, Instructor, Course, Module, Assignment, Submission, Grade, Feedback, LearningObjective, LateSubmission, Prerequisite, Enrollment **Terms from Answers:** enrolledIn, teaches, hasModule, hasAssignment, submits, submittedBy, submissionFor, hasGrade, hasFeedback, hasLearningObjective, hasPrerequisite **Objects:** • Person → Student, Instructor • AcademicContent → Course, Module, LearningObjective • Assessment → Assignment • Submission → separate semantic class • Evaluation → Grade, Feedback
9	**Ontology Usage and**
	The ontology was extended during Phase 2 using ontology engineering principles inspired by METHONTOLOGY and Modular Ontology Modeling approaches. **New Classes Added:** • LateSubmission • LearningObjective

	• Feedback **New Object Properties Added:** • hasLearningObjective • hasFeedback • isAssignmentOf • isModuleOf • isLearningObjectiveOf **New Datatype Properties Added:** • moduleTitle • moduleOrder • submissionDate • dueDate **Semantic Enhancements:** • Inverse properties • Functional properties • Disjointness axioms • Transitive prerequisite reasoning • Automatic LateSubmission classification • Consistency validation using HermiT
10	**Research Integration**
	The selected research integration topic for Phase 2 was Ontology Population using Large Language Models (LLMs). The ontology structure was designed to support future automated extraction of educational entities and relationships from: • LMS logs • Course catalogues • Assignment briefs • Gradebooks • Instructor profiles The ontology population pipeline concept includes: 1. Schema-aware prompting 2. Educational entity extraction 3. RDF/OWL assertion generation 4. Reasoner-based validation The project was also influenced by the mandatory reviewed paper: “Personalized Ontology and Deep Training Tree-Based Optimal GRU-RNN for Prediction of Students’ Behavior”, which inspired future directions related to intelligent educational analytics and semantic student modeling.

11	**Version 2 Additions**
	The following major additions were integrated into ORSD Version 2: • Added semantic classes: - Feedback - LearningObjective - LateSubmission • Added new competency questions related to: - learning objectives - feedback - late submissions - reasoning • Added ontology population concepts using LLMs. • Added reasoning-related semantic enhancements: - inverse properties - transitive prerequisite inference - disjointness axioms - functional properties • Added modular ontology design concepts. • Added support for intelligent educational analytics and future AI integration.